

## UNIT – II DIELECTRIC PROPERTIES

### Introduction

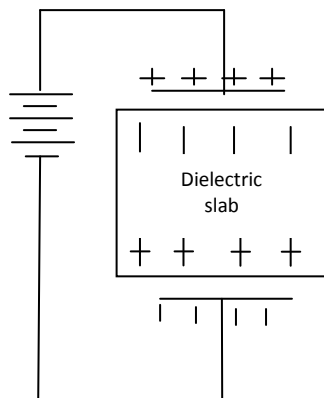
Solids which have an energy gap of three eV or more are termed as insulators. In these materials, it is almost not possible to excite the electrons from the valence band to conduction band by an applied field. Generally dielectrics are also called as insulators, thereby poor conductors of electricity. However, they allow movement of some electrons at abnormally high temperatures, causing a small flow of current.

Dielectrics are non-metallic materials of high specific,  $\rho$  resistance, negative temperature coefficient of resistance ( $-\alpha$ ) and large insulation resistance. Insulation resistance will be affected by moisture, temperature, applied electric field and age of dielectrics.

### Fundamental definitions and properties

#### 1. Electric polarization

When an external electric field is applied to the dielectrics, the field exerts a force on each positive charges in its own direction while negative charges are pushed in the opposite directions.



Fig(1)

Consequently, an electrical doublet or dipole is created in all the atoms inside the dielectric as shown in the above figure (1).

Thus, the process of producing electric dipoles inside the dielectrics by an external electric field is called polarization in dielectrics.

## 2. Polarization vector( $\vec{p}$ )

If the strength of electric field  $E$  is increased, the strength of the induced dipoles also increases. The induced dipole moment is proportional to the intensity of the electric field.

$$\mu = \alpha E$$

Where  $\alpha$  is the constant of proportionality is called the polarizability.

If  $\mu$  is the average dipole moment per molecule and  $N$  is the number of molecules per unit volume, the polarization vector  $\vec{P}$  is defined as dipole moment per unit volume of the dielectric material

$$\vec{P} = N\mu$$

## 3. Electric displacement vector( $\vec{D}$ )

Electric displacement vector (or) electric induction  $\vec{D}$  is a quantity which is used for analyzing electrostatic fields in the presence of dielectrics which is given by

$$\vec{D} = \frac{Q}{4\pi r^2} \text{ --- } \rightarrow \quad (1)$$

We know electric field intensity

$$\vec{E} = \frac{Q}{4\pi\epsilon r^2} \text{ ----> } \quad (2)$$

Form (1) and (2) we can write

$$\begin{aligned} \vec{D} &= \epsilon \vec{E} \\ &= \epsilon_0 \epsilon_r \vec{E} \text{ ----> } \end{aligned} \quad (3)$$

$$\vec{D} = \epsilon_0(1+\chi_e)\vec{E} \text{ [since } \epsilon_r=1+\chi_e]$$

Where  $\chi_e$  is the electrical susceptibility

$$\vec{D} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E}$$

Since  $\vec{P} = \epsilon_0 \chi_e \vec{E}$ , we have

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \text{ -----> } \quad (4)$$

Equating (3) and (4) we have

$$\epsilon_0 \epsilon_r \vec{E} = \epsilon_0 \vec{E} + \vec{P}$$

$$\frac{\vec{P}}{\vec{E}} = \epsilon_0(\epsilon_r - 1) \text{ ----> } \quad (5)$$

#### 4. Electrical susceptibility( $\chi_e$ )

The polarization vector  $\vec{P}$  is proportional to the applied electric field  $\vec{E}$ , for field strength that are not too large so we can write

$$\vec{P} \propto \vec{E} \quad \text{---} \rightarrow \quad (1)$$

$$(\text{or}) \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$\chi_e$  is a characteristic of every dielectric and which is called electric susceptibility

$$\chi_e = \frac{\vec{P}}{\epsilon_0 \vec{E}}$$

Since  $\frac{\vec{P}}{\vec{E}} = \epsilon_0(\epsilon_r - 1)$  We can write

$$\chi_e = \frac{(\epsilon_r - 1)\epsilon_0}{\epsilon_0}$$

$$\chi_e = \epsilon_r - 1 \quad \text{---} \rightarrow \quad (2)$$

## 5. Dielectric constant( $\epsilon_r$ )

The dielectric constant determines the share of the dielectric stress which is absorbed by the material. It is the ratio between the absolute permittivity  $\epsilon$  and the permittivity of free space  $\epsilon_0$  and is given by

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

$\epsilon_r$  is a dimensionless quantity and it is a measure of polarization in the dielectrics.

The value of  $\epsilon_r = 1$  for air or vacuum.

## 6. Active dielectrics

When dielectric is subjected to external electric field, if the dielectric actively accept the electricity then they are termed as active dielectrics. Thus

active dielectrics are which can easily adapt itself to store the electrical energy in it.

**Example:** Piezo-electrics, Ferro-electrics

## 7. Passive dielectrics

Passive dielectrics are which restricts the flow of electrical energy in it.

**Example :** glass, mica, etc...

## 8. Polar and non-polar molecules

Polar molecules are the molecules in which centre of negative and positive charges will not coincide and hence it possess a net dipole moment in it.

Examples:  $\text{H}_2\text{O}$ ,  $\text{N}_2\text{O}$ ,  $\text{HCl}$ ,  $\text{NH}_3$ ,  $\text{CO}$ ,  $\text{CH}_3\text{OH}$ , etc...

Non-polar molecules are the molecules in which the centre of negative and positive charges coincide and hence they do not possess a net dipole moment in it.

Example:  $\text{N}_2$ ,  $\text{H}_2$ ,  $\text{O}_2$ ,  $\text{CH}_4$ ,  $\text{CO}_2$ , etc.,

### Difference between polar and non-polar molecules

| Sl. No. | Polar molecules                                                                       | Non-Polar molecules                                                      |
|---------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1       | These molecules have permanent dipole moments even in the absence of an applied field | These molecules do not have permanent dipole moments.                    |
| 2       | The polarization of polar molecules is highly temperature dependent                   | The polarization of this kind of molecules is independent of temperature |
| 3       | These molecules do not have                                                           | These molecules have symmetrical                                         |

|   |                                                                                    |                                                                                 |
|---|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|   | symmetrical structure and they do not have centre of symmetry                      | Structure and they have centre of symmetry                                      |
| 4 | For this kind of molecules, there is absorption or emission in the infrared range. | For these molecules there is no absorption or emission in the range of infrared |
| 5 | Eg: CH <sub>3</sub> Cl, HCl, H <sub>2</sub> O                                      | Eg: CCl <sub>4</sub> , CO <sub>2</sub> , H <sub>2</sub>                         |

### Various polarization mechanism in dielectrics

#### Dielectric polarization:

It is the displacement of charged particles under the action of the external electric field. There are number of devices based on this concept. Those devices are rectifiers, resonators, amplifiers and transducers, which converts electrical energy to other forms of energy.

#### In modern computers memory devices are also based on this concept.

Several microscopic mechanisms are responsible for electric polarization. Specially in the case of d.c. electric field, the macroscopic polarization vector  $\vec{P}$  arises due to the following four types of microscopic polarization mechanisms.

1. Electronic polarization ( $\alpha_e$ )
2. Ionic polarization ( $\alpha_i$ )
3. Orientation polarization ( $\alpha_o$ )
4. Space-charge polarization

## Electric polarization:

Electronic polarization occurs due to the displacement of positively charged nucleus and negative charged electrons in opposite directions when an external electric field is applied and thereby creates a dipole moment in the dielectric.

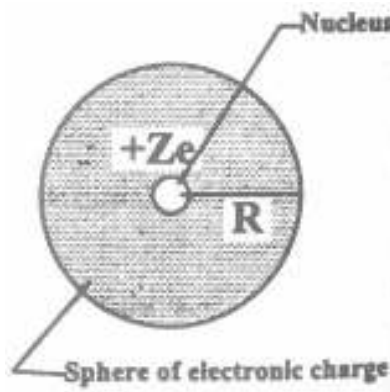
The induced dipole moment  $\mu = \alpha_e E$  Where  $\alpha_e$  electronic polarizability

Mono atomic gases exhibit this kind of polarization. Electronic polarizability is proportional to the volume of the atoms and is independent of temperature.

### Calculation of Electronic polarizability

#### (i) Without field

Let us consider a classical model of an atom. Assume the charge of nucleus of that atom is  $+Ze$ . The nucleus is surrounded by an electron cloud of charge  $-Ze$ . Which is distributed in a sphere of radius  $R$  as shown below



The charge density of the charged sphere =  $\frac{-Ze}{\frac{4}{3}\pi R^3}$

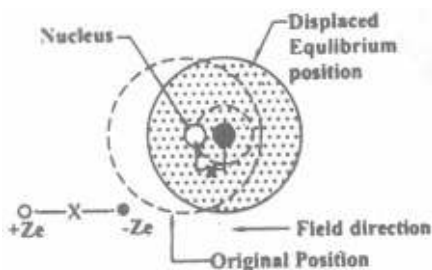
$$\text{Or charge density} = \frac{-3Ze}{4\pi R^3} \rightarrow \quad (1)$$

#### (ii) with field

When the dielectric is placed in a d c electric field  $E$ , two phenomenon occur.

(i) Lorentz force due to the electric field  $E$ , tends to separate the nucleus and the electron cloud from their equilibrium position.

(ii) After separation, an attractive coulomb force arises between the nucleus and electron cloud which tries to maintain the original equilibrium position. Let 'x' be the displacement made by the electron cloud from positive core as shown below.



Since the core is heavy, it will not move when compared to the movement of electron cloud. Here  $x \ll R$  where 'R' is the radius of atom.

Since Lorentz and coulomb forces are equal and opposite in nature, equilibrium is reached at

At Equilibrium

Lorentz force = coulomb force

Lorentz force = charge  $\times$  field

$$= -ZeE \quad \text{---} \rightarrow \quad (2)$$

$$\text{Coulomb force} = \frac{\text{Charge of } x \text{ total negative charge nucleus enclosed in the space of radius } x}{4\pi\epsilon_0 x^2} \rightarrow (3)$$

Here, total number of negative charges

enclosed in the sphere of radius  $x$  = charge density of electrons  $\times$  volume of sphere of radius.

Substituting the charge density from equation (1) we get



Total number of negative charges enclosed in the sphere of radius  $x = \frac{-3 Ze}{4 \pi R^3} \times \frac{4}{3} \pi x^3$

$$= \frac{-Zex^3}{R^3} \longrightarrow (4)$$

Substituting equation (4) in (3) we get

$$\text{Coulomb force} = \frac{Zex}{4\pi\epsilon_0 x^2} \left[ \frac{-Zex^3}{R^3} \right]$$

$$= \frac{-Z^2 e^2 x}{4\pi\epsilon_0 R^3} \longrightarrow (5)$$

At equilibrium position eq (2) eq (5)

$$-ZeE = \frac{-Z^2 e^2 x}{4\pi\epsilon_0 R^3}$$

$$E = \frac{Zex}{4\pi\epsilon_0 R^3}$$

$$x = \frac{4\pi\epsilon_0 R^3 E}{Ze} \longrightarrow (6)$$

therefore the displacement of electron cloud ( $x$ ) is proportional to applied field  $E$

Now the two electric charges  $+ze$  and  $-ze$  are displaced by a distance  $x$  under the influence of the field and from an induced dipole moment which is given by

Induced dipole moment  $\mu_e = \text{magnitude of charge} \times \text{displacement}$

Induced dipole moment ( $\mu_e$ ) =  $zex$

substituting the value of  $x$  from eq (6) we have  $\mu_e = \frac{Ze4\pi\epsilon_0 R^3 E}{Ze}$

$$\mu_e = 4\pi\epsilon_0 R^3 E$$

or

$$\mu_e \propto E$$

$$(\text{or}) \mu_e = \alpha_e E \quad \rightarrow \quad (7)$$

where  $\alpha_e = 4\pi\epsilon_0 R^3$  is called electronic polarization which is proportional to volume of the atom.

### Relation between $\alpha_e$ and dielectric constants

We know, that induced electronic dipole moment is proportional to the applied field. This dipole moment per unit volume is called electronic polarization this is independent of temperature

$$\text{Electronic polarization } \vec{P}_e = N\mu_e$$

$$\text{Or} \quad \vec{P}_e = N\alpha_e \vec{E} \quad (8)$$

Where N is the number of atoms

Since  $N\alpha_e = \epsilon_0(\epsilon_r - 1)$  we get

$$\vec{P}_e = \vec{E} \epsilon_0(\epsilon_r - 1) \rightarrow (9)$$

Equating equations (8) and (9) we get

$$N\alpha_e \vec{E} = \vec{E} \epsilon_0(\epsilon_r - 1)$$

$$\alpha_e = \frac{\epsilon_0(\epsilon_r - 1)}{N} \rightarrow (10)$$

## Ionic polarisation:

Ionic polarisation arises due to the displacement of cations (+ve ion and anions (-ve ions) from its original position in opposite direction in the presence of electric field as shown in the below figure

The displacement is independent of temperature and it occurs in ionic solids.

**Eg** NaCl crystals.



**Explanation:** let us assume that they are one cation and one anion present in each unit cell of the ionic crystal when the electric field is applied, let  $x_1$  and  $x_2$  be the distances to which positive and negative ions move from their equilibrium positions. The resultant dipole moment per unit cell, due to Ionic displacement is given by

$$\mu_1 = e(x_1 + x_2) \text{ -----} \blacktriangleright \quad (1)$$

Where  $x_1$  is the shift of +ve ion and

$x_2$  is the shift of -ve ion, from their equilibrium positions.

When the field is applied, the restoring - force produced is proportional to the displacement

For +ve ion restoring force  $F \propto x_1$

$$F = \beta_1 x_1 \text{ -----} \blacktriangleright \quad (2)$$

$$F \propto x_2$$

$$F = \beta_2 x_2 \quad \text{-----} \rightarrow \quad (3)$$

Here  $\beta_1$  and  $\beta_2$  are restoring force constants, which depend on the masses of the ions and the angular frequency of the molecule in which ions are present.

If  $m$  is the mass of +ve ions and  $M$  is the mass of -ve ion and  $\omega_0$  is the angular frequency, then

$$\beta_1 = m\omega_0^2 \quad \text{-----} \rightarrow \quad (4)$$

$$\beta_2 = M\omega_0^2 \quad \text{-----} \rightarrow \quad (5)$$

Where  $\omega_0$  -angular frequency

Substituting for  $\beta_1$  in eq (2) the restoring force for +ve ion can be written as

$$F = m\omega_0^2 x_1 \quad \text{-----} \rightarrow \quad (6)$$

$$\text{We know } F = eE \quad \text{-----} \rightarrow \quad (7)$$

Equating eq (6) and eq (7) we get  $eE = m\omega_0^2 x_1$

$$x_1 = \frac{eE}{m\omega_0^2} \quad \text{-----} \rightarrow \quad (8)$$

Similarly for the negative ion we can write

$$X_2 = \frac{eE}{M\omega_0^2} \quad (9)$$

Adding eq (8) and (9) we get

$$X_1 + X_2 = \frac{eE}{\omega_0^2} \left( \frac{1}{m} + \frac{1}{M} \right) \text{-----} \rightarrow \quad (10)$$

Subtracting equation (10) in (1) we get

$$\mu = \frac{eE}{\omega_0^2} \left( \frac{1}{m} + \frac{1}{M} \right)$$

or  $\mu_i = \alpha_i E$

where  $\alpha_i$ - ionic polarizability, given by

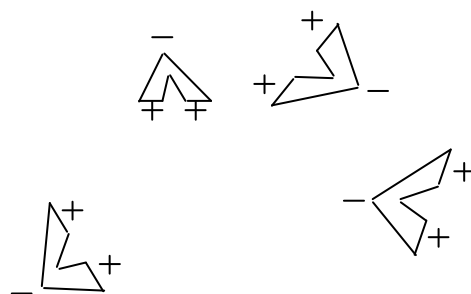
$$\alpha_i = \frac{e^2}{\omega_0^2} \left( \frac{1}{m} + \frac{1}{M} \right)$$

So the ionic polarizability  $\alpha_i$  is inversely proportional to the square of the natural frequency of the ionic molecules and directly proportional to its reduced mass is given

by  $-\left( \frac{1}{m} + \frac{1}{M} \right)$

### Orientation polarization

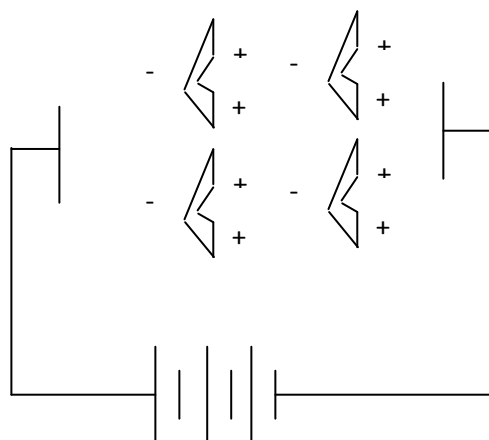
Polar molecules are the which have permanent dipole moments even in the absence of an electric field as shown below.



Without field

The orientation polarization arises due to the presence of polar molecule in the dielectric medium. When a dielectric which consists of polar molecules is kept in an

electric field, the molecules align themselves along the field direction. So there is a resultant dipole moment along the field direction as shown below.



With field

**Explanation:** in the case of a  $\text{CH}_3\text{Cl}$  molecule the +ve and -ve charges do not coincide. The  $\text{Cl}^-$  has more electro negativity then hydrogen. Therefore the chlorine atoms pull the bonded electrons towards it more strongly than hydrogen atoms. Therefore even in the absence of field, there exists a net dipole moment.

Now, when the field is applied, positive portion align along the direction of field and negative portion align in the opposite direction of the field. This kind of polarization is called as orientation polarization .This depends on temperature. When temperature is increased, the thermal energy tends to randomize the alignment.

From langevins theory of paramagnetism net intensity of magnetization

$$\text{Intensity of magnetisation} = \frac{N\mu^2/3}{3K_B T}$$

Since, the same principle can be applied to the application of electric field

$$\text{orientation polarization } P_o = \frac{N\mu^2 E}{3k_b T}$$

where  $N$  is the number of atoms

$$(or) P_o = N\alpha_o E$$

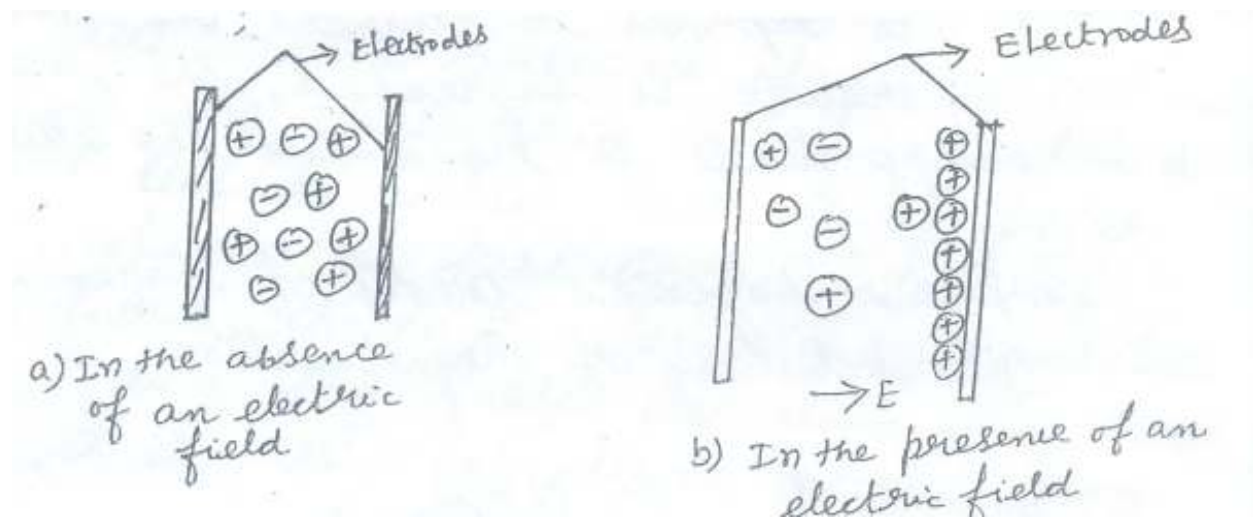
Where  $\alpha_o$  orientational polarizability

$$\alpha_o = \frac{\mu^2}{3k_B T}$$

Therefore orientational polarizability is inversely proportional to the temperature of the material

### Space – Charge polarization

The space-charge polarization occurs due to diffusion of ions, along the field direction and giving rise to redistribution of charges in the dielectrics



### Explanation:

Without the application of external field, the ions are orderly arranged as shown above.

Now when the field is applied the ions diffuse with respect to the direction of applied field. Thus the polarization occurs known as space charge polarization

Normally this type of polarization occurs in ferrites and semiconductors and will be very small

### **Total Electric polarization:**

The total electric polarization is the sum of electronic polarization, ionic polarization, orientation polarization and space charge polarization. Since space charge polarization is very small when compared to other kinds of polarization it can be neglected. Therefore the total polarizability is given by

$$\alpha = \alpha_e + \alpha_i + \alpha_o$$

$$\alpha = 4\pi\epsilon_0 R^3 + \frac{e^2}{\omega_0^2} \left( \frac{1}{m} + \frac{1}{M} \right) + \frac{\mu^2}{3k_B T}$$

We know total polarization  $P = NE \alpha$

$$P = NE \left[ 4\pi\epsilon_0 R^3 + \frac{e^2}{\omega_0^2} \left( \frac{1}{m} + \frac{1}{M} \right) + \frac{\mu^2}{3k_B T} \right]$$

This equation is called as Langevin Debye equation

### **Measurement of dielectric constant**

We know total polarization  $P = \text{Electronic} + \text{ionic} + \text{orientation polarization}$

$$P = N\alpha_e E + N\alpha_i E + N\alpha_o E$$

$$P = NE \left( \alpha_e + \alpha_i + \frac{\mu^2}{3k_B T} \right) \text{ -----} \rightarrow \quad (1)$$

Since  $P = E\epsilon_0 (\epsilon_r - 1)$  and  $\alpha = \alpha_e + \alpha_i$

Equation (1) can be written as



$$P = E\epsilon_0 (\epsilon_r - 1) = NE \left( \alpha^1 + \frac{\mu^2}{3k_bT} \right)$$

Since  $\chi_e = \epsilon_r - 1$  we can write

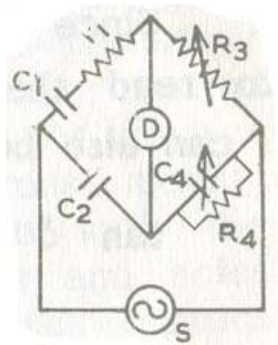
$$\epsilon_r - 1 = \chi_e = \frac{N}{\epsilon_0} \left( \alpha^1 + \frac{\mu^2}{3k_bT} \right)$$

$$(\text{or}) \epsilon_r = \frac{N}{\epsilon_0} \left( \alpha^1 + \frac{\mu^2}{3k_bT} \right) + 1$$

### Measurement of dielectric constant and dielectric loss of the dielectric using Schering-bridge:

It is an experimental method used for the accurate measurement of small capacitances.

The connections are shown in the below figure:



The unknown capacitor  $c$  is connected in arm AB along with the associated resistance  $R_1$  in series. The arm AD contains a standard capacitor  $C_3$  having negligible resistance. The two remaining arms BC and CD contain non-inductive resistances  $R_2$  and  $R_4$ . A variable calibrated air capacitor  $C_4$  is put in parallel with  $R_4$ . The bridge is balanced by varying  $R_2$  and  $C_4$ . First, without dielectric slab inside  $c$ , balance is obtained by adjusting  $C_4$  and  $R_2$ .

The general balance condition of an a.c bridge is

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$

We know that impedance in arm 1 and 2 are  $Z_1 = R_1 + \frac{1}{j\omega C}$  and  $Z_2 = R_2$

In arm (4) the capacitor  $C_4$  and  $R_4$  are parallel therefore the reactive capacitance and impedances in arm (4) is

$$\frac{1}{Z_4} = \frac{1}{R_4} + \frac{1}{\frac{1}{j\omega C_4}}$$

In arm (3) the reactive capacitance of capacitor  $C_3$  is  $Z_3 = \frac{1}{j\omega C_3}$

Using null deflection condition

$$\frac{R_1 + \frac{1}{j\omega C}}{R_2} = \frac{1}{j\omega C_3} + \left( \frac{1}{R_4} + j\omega C_4 \right)$$

$$R_1 + \frac{1}{j\omega C} = \frac{R_2}{j\omega C_3 R_4} + \frac{R_2 C_4}{C_3}$$

Equating real and imaginary parts

$$R_1 = R_2 \frac{C_4}{C_3} \quad \text{-----} \rightarrow \quad (1)$$

and

$$\frac{1}{C} = \frac{R_2}{C_3 R_4}$$

$$C = \frac{R_4}{R_2} C_3 \quad \text{-----} \rightarrow \quad (2)$$

Equation (2) gives the value of the capacitance of free space (ie) without specimen

Now with the dielectric slab whose dielectric constant has to be measured is inserted between the plates C. It must be remembered that the area of the plates must be equal to the area of the slab and also the thickness of the slab must be equal to the air gap between the electrodes.

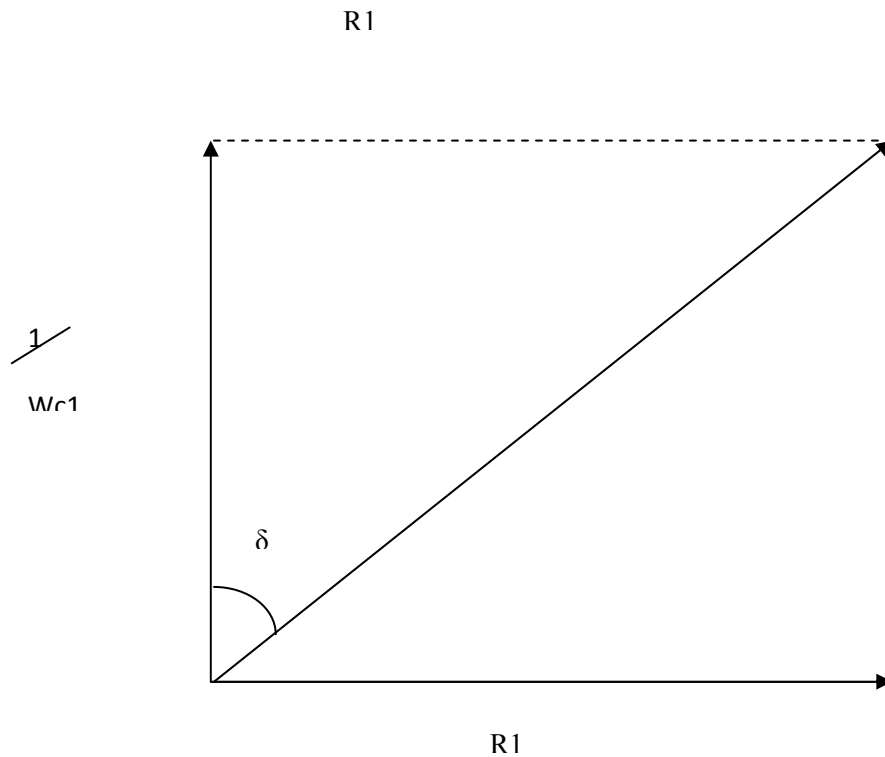
Now with the slab inside,  $R_2$  is dialed to  $R_2$  to get the bridge balanced and the capacitance  $C_{sp}$  is calculated

$$C_{sp} = \frac{R_4}{R_2} C_3$$

Now the dielectric constant of the specimen

$$\epsilon_r = \frac{C_{sp}}{C_{air}}$$

From the phasor diagram drawn for capacitive reactance



Loss tangent tan

$$\delta = \frac{r_1}{\omega C_{sp}}$$

$$\tan \delta = \omega C_{sp} r_1$$

Substitution the values of  $r_1$  and  $C_{sp}$  in the above equation we get

$$\tan \delta = \omega \times \left( \frac{R_4}{R_2} \right) C_3 \times \frac{R_2 C_4}{C_3}$$

$$\tan \delta = \omega R_4 C_4 \left( \frac{R_2}{R_2} \right)$$

Thus one can measure the dielectric constant and loss tangent experimentally. One can calculate the dielectric constant at different frequencies by changing the input frequency of the transformer.

### Frequency and temperature dependence of all the polarization mechanisms

When field is applied, the polarization occurs as a function of time. The polarization  $P(t)$  as a function of time  $t$  is given by

$$P(t) = P[1 - e^{(-t/t_r)}]$$

Where  $P$  is the maximum polarization which occurs at the static field application for a long time and  $t_r$  is the relaxation time (i.e) the time taken for polarization. It is a measure of the time scale of polarization process.

Relaxation time is the time taken for the polarization process to reach 0.63 of the maximum value of polarization

The relaxation times are different for different kinds of polarization mechanisms.

### a) Frequency dependence

- (i) Electronic polarization is very very rapid and will complete at the instant the voltage is applied the reason is that the electrons are very. Very light elementary particles than ions. Therefore even for high frequency applied voltage, ie, in the optical range  $\approx 10^{15}$  hz as shown in the below figure.

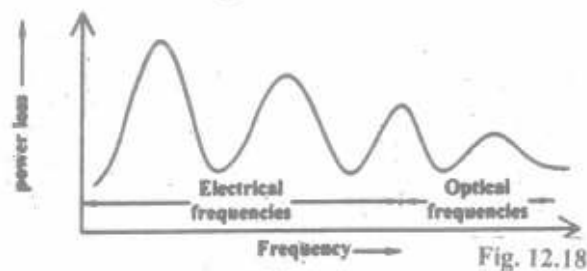
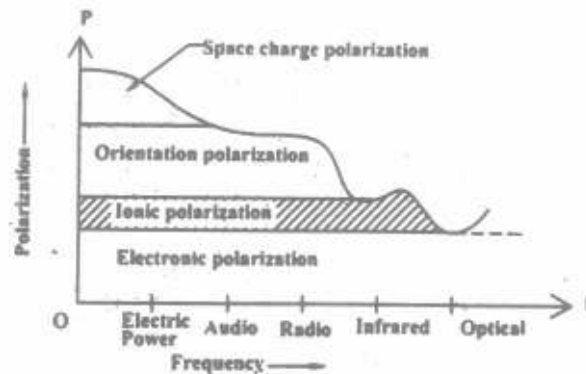


Fig. 12.18

- (i) Ionic polarization is slightly slower than the electronic polarization. Because ions are heavier than the electron cloud. Also the frequency of the applied electric field with which the ions will be displaced is equal to the frequency of the lattice vibrations ( $\sim 10^{13}$  Hz)

At optical frequency there is no ionic polarization. If the frequency of the applied voltage is less than  $\sim 10^{13}$  Hz (i.e) infrared range as shown above the ions have enough time to respond during each cycle of the applied field.

- (ii) Orientation polarization is even slower than ionic polarization. The relaxation time for this case varies with respect to the dielectric materials (

ie, solids or liquids) used here the polar molecules in a liquid easily reorient themselves compared to solids. This type of polarizations occurs at radio frequency ranges  $\sim 10^6$  Hz as shown above

- (iii) Space charge polarization is the slowest process, because in this case the ions have to diffuse (jump) over several inter atomic distances. Also this process occurs at very low frequency in the order of  $10^2$  hz as shown.

Therefore from the figure we can observe that, at lower frequencies all the four types of polarization occurs and the total polarization is maximum. And the total polarization value decreases with the increase in frequency and becomes minimum at optical frequency range

#### **b) Temperature dependence:**

The electronic and ionic polarizations are independent of temperature, whereas the orientation and space charge polarizations are temperature dependent.

The orientation polarization decreases with the increase in temperature because the randomizing action of thermal energy decreases the tendency of the permanent dipoles to align along the field direction Hence in this case the  $\epsilon_r$  decreases.

But in space charge polarization when the temperature is increased, the ions can easily overcome the activation barrier and hence they diffuse through the inter atomic distances. Thus it gives rise to polarization. So in this case the  $\epsilon_r$  will increase with the increase in temperature.

#### **Internal field (or) local field and deduction of CLAUSIUS – MOSOTTI Equation**

When a dielectric is kept in an external field it exerts a dipole moment in it. Therefore two fields are exerted, viz.

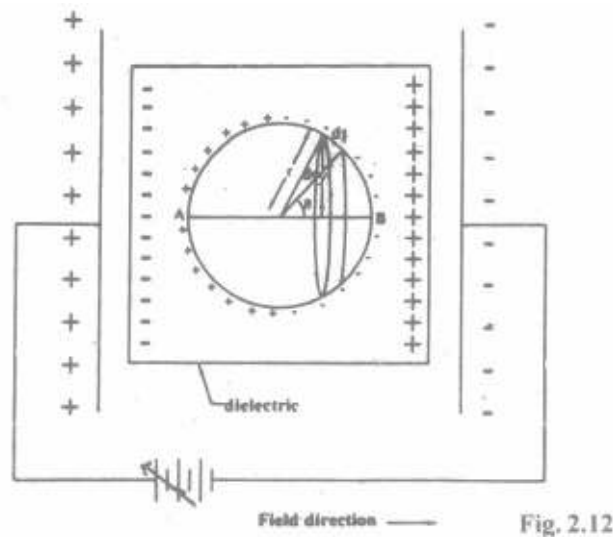
- (i) Due to external field.

(ii) Due to dipole moment

This long range of coulomb forces which is created due to the dipoles are called as internal field or local field. This field is responsible for polarising the individual atoms or molecules.

**Lorentz method for finding Internal field.**

Let us assume, we are placing a dielectric material in an external field. Consider an imaginary sphere in the solid dielectric of radius 'r' as shown in the diagram.



Here the radius of the sphere is greater than the radius of the atoms. ie there are many atomic dipoles within the sphere. A small elemental ring is cut with thickness  $d_s$ . Let  $y$  be the radius of the small ring as shown below.

The electric field at the centre of the sphere is called internal field. Which arises due to the following four factors.

$$E_{\text{int}} = E_1 + E_2 + E_3 + E_4 \quad \text{-----} \rightarrow \quad (1)$$

Where

$E_1$  field due to the charge on the plates (externally applied)

$E_2$  field due to polarization charges on the plane surface of the dielectric

$E_3$  field due to polarized charges induced at the spherical surface.

$E_4$  field due to atomic dipoles inside the sphere considered.

Macroscopically, we can take  $E_1 = E_1 + E_2$  (i.e) the field externally applied ( $E_1$ ) and the field induced on the plane surface of the dielectric ( $E_2$ ) as a single field ( $E$ )

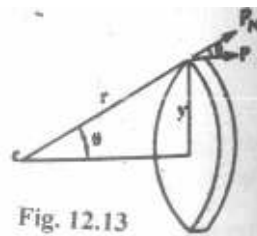
If the dielectric is highly symmetric then the dipoles will cancel with each other therefore we can take  $E_4 = 0$

Equation 1 becomes

$$E_{\text{int}} = E + E_3 \longrightarrow (2)$$

To find  $E_3$

In the elemental ring let  $q'$  be the charge on the area  $d_s$ . Polarization is defined as the surface charges per unit area. If  $P_N$  is the component of polarization perpendicular to the area as shown below.



$$\text{Here } P_N = P \cos \theta = \frac{q'}{d_s}$$

$$(\text{or}) Q' = P \cos \theta d_s$$

Electric field intensity at c due to charge  $q'$  is given by



$$E = \frac{q}{4\pi\epsilon_0 r^2} \rightarrow \quad (3)$$



The above intensity is along the radius 'r' Resolving the intensity into two components as shown below.

Component parallel to the field direction  $E_x = E \cos \theta$

$$E_x = \frac{P \cos^2 \theta ds}{4\pi\epsilon_0 R^2}$$

Component perpendicular to the field direction  $E_y = E \sin \theta$

$$E_y = \frac{P \cos \theta \sin \theta ds}{4\pi\epsilon_0 r^2}$$

The perpendicular components are in opposite directions and hence cancel each other. So the parallel components are alone taken into consideration.

If the total surface area of the ring considered as  $dA$  then

$$E_x = E = \frac{P \cos^2 \theta dA}{4\pi\epsilon_0 r^2}$$

Where  $dA$  Is the surface area of the elemental ring.

$dA = \text{circumference} \times \text{thickness}$

$$= 2\pi y x r d_o \text{ (are length)}$$

$$\text{from the figure } \sin \theta = \frac{y}{r}$$

$$Y = r \sin \theta$$

$$dA = 2\pi r^2 \sin \theta d\theta \rightarrow (5)$$

sub eg (5) in (4) we get electric field intensity due to the elemental ring

$$= \frac{P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0} \rightarrow (6)$$

electrical field intensity due to the whole sphere can be derived by integrating equation within the limits 0 to  $\pi$

$$E_3 = \int_0^\pi \frac{P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0}$$

W.K.T

$$E_3 = \frac{P}{2\epsilon_0} \int_0^\pi \cos^2 \theta d(-\cos \theta)$$

$$\text{Let } x = \cos \theta$$

$$E_3 = \frac{-P}{2\epsilon_0} \int_0^\pi x^2 dx = \frac{-P}{2\epsilon_0} \left[ \frac{x^3}{3} \right]$$

$$E_3 = \frac{-P}{2\epsilon_0} \left[ \frac{\cos^3 \theta}{3} \right]$$

$$E_3 = \frac{-P}{2\epsilon_0} \left[ \frac{(\cos \pi^3) - (\cos \pi^3)}{3} \right]$$

$$E_3 = \frac{-P}{2\epsilon_0} [(-1) - 1]$$

Substituting eq (7) in eq( 2) we can write

$$E_{\text{int}} = E + P/3\epsilon_0$$

Where  $E_{\text{int}}$  is called infernal field (or) lorintz field Substituting eq 9 in equ 8 we get

### CLAUSIUS MOSOTTI EQUATION

We know  $p = \epsilon E = \epsilon_0 E + P$

$$E(\epsilon - \epsilon_0) = P$$

$$E = P / (\epsilon - \epsilon_0) \rightarrow (9)$$

Substituting eq (9) in eq (8) we get

$$E_{\text{int}} = \frac{P}{\epsilon - \epsilon_0} + \frac{P}{3\epsilon_0}$$

$$= \frac{P[3\epsilon_0 + (\epsilon - 3\epsilon_0)]}{3\epsilon_0(\epsilon - \epsilon_0)}$$

$$E_{\text{int}} = \frac{P(2\epsilon_0 + \epsilon)}{3\epsilon_0(\epsilon - \epsilon_0)} \rightarrow (10)$$

Since there are no ions and permanent dipoles in the dielectric materials, the ionic polarizability  $\alpha_i$  and orientation polarizability  $\alpha_o$  are zero.

Therefore Net Polarization  $P = N\alpha_e E_i$

$$E_{\text{int}} = \frac{P}{N\alpha} \text{-----} \rightarrow (11)$$

Comparing eq (10) and eq (11) we get

$$\frac{P}{N\alpha} = \frac{P(2\varepsilon_o + \varepsilon)}{3\varepsilon_o(\varepsilon - \varepsilon_o)}$$

$$\frac{3\varepsilon_o}{N\alpha} = \frac{P(2\varepsilon_o + \varepsilon)}{p(\varepsilon - \varepsilon_o)}$$

$$\frac{N\alpha}{3\varepsilon_o} = \frac{\varepsilon - \varepsilon_o}{2\varepsilon_o + \varepsilon}$$

$$\frac{N\alpha}{3\varepsilon_o} = \frac{\varepsilon_o(\varepsilon/\varepsilon_o - 1)}{\varepsilon_o(\frac{\varepsilon}{\varepsilon_o} + 2)}$$

W.K.T

$$\frac{\varepsilon}{\varepsilon_o} = \varepsilon_r$$

$$\frac{N\alpha}{3\varepsilon_o} = \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \text{-----}(12)$$

Equation (12) represents clausius mossotti relation

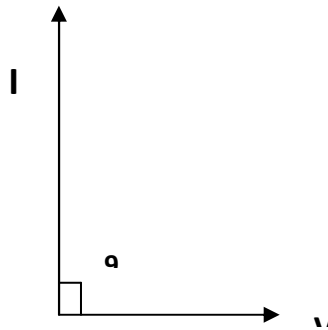
### Dielectric loss

If a dielectric is subjected to an electric field, the electrical energy is absorbed by the dielectric and certain quantity of electrical energy is dissipated in the form of heat energy. This is known as dielectric loss.

The dielectric loss can occur both in direct and alternating voltages. The dielectric loss is less in direct voltage than that of alternating voltages.

### Loss in purified gas

If an alternating voltage is applied across the capacitor having vacuum (or) purified gas then the resulting current leads the applied voltage by 90° as shown below. If 'I' leads V exactly by 90° we can say that no electrical energy is lost.



### Explanation

We know power loss  $P_L = VI \cos \theta$

When  $\theta = 90^\circ$

$$P_L = 0$$

### Loss in commercial dielectric

Now, when a practical dielectric is present the current leads the voltage by  $(90^\circ - \delta)$  then it shows that there is some loss in electrical energy and  $\delta$  is called loss angle as shown below.

### Explanation

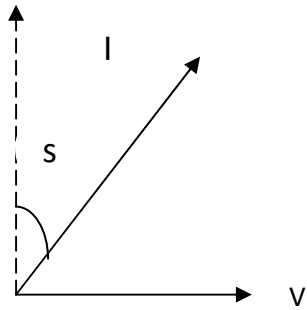
In this case the power loss  $P_L = VI \cos \theta$

Since  $\theta = 90^\circ$  we have  $P_L = VI \cos(90^\circ - \delta)$

$$P_L = vI \sin \delta \quad (1)$$

we know  $v = IR$

$$I = V/R$$



If the capacitive reactance is  $X_c$  then we can write

$$I = v/x_c \quad (2)$$

Substituting eq (2) in eq (1) we get

Power loss  $P_L$

$$\frac{v^2 \sin \theta}{x_c} \text{-----} (3)$$

We know frequency  $f = \frac{1}{2\pi RC}$

$$= \frac{1}{2\pi x_c C}$$

$$X_c = \frac{1}{2\pi f C}$$

Substituting eq (4) in eq(3) we get

$$P_L = 2\pi f c V^2 \sin \theta$$

If  $\delta$  is very small then  $\sin \delta = \tan \delta$

$$\text{Power loss } P_L = 2\pi f c V^2 \sin \theta$$

Here  $\tan \delta$   $P_L$  is called the power factor of the dielectric If  $f, c, v$  are constants

$$P_L \propto \tan \delta$$

Naturally the power loss varies with frequency.

In the radio frequency region the power loss is high, due to the diffusion of ions from equilibrium position to another.

In the optical region the power loss is less because here the dielectric loss is associated with the electrons.

### **Dielectric breakdown**

When a dielectric material loses its resistivity and permits very large current to flow through it, then the phenomenon is called dielectric breakdown. The different types of dielectric breakdown are

- 1 Intrinsic breakdown (or) zener breakdown
- 2 Thermal breakdown.
- 3 Discharge breakdown
- 4 Electro chemical breakdown.
- 5 Defect breakdown.
- 6 Avalanche breakdown.

### **1. Intrinsic breakdown:**

When the applied electric field is large, some of the electrons in the valence band cross over to the conduction band across the large forbidden energy gap giving rise to large conduction currents. The liberation or movement of electrons from valence band is called field emission of electrons and the breakdown is called intrinsic breakdown or zener breakdown

### **2. Avalanche breakdown:**

If the electric fields is of the order of  $10^6$  v/cm the electrons are accelerated to very high velocity colliding with the other atoms and molecules, thereby releasing more electrons and holes breaking covalent bonds between them. The number of bonds broken and hence number of charge carriers released increases enormously with time and finally dielectric breakdown occurs. This type of breakdown is called avalanche breakdown.

### **3. Thermal breakdown:**

Thermal breakdown occurs in a dielectric, when the rate of heat generation is greater than the rate of dissipation. Energy due to the dielectric loss appears as heat. If the rate of generation of heat is larger than the heat dissipated to the surroundings, the temperature of the dielectric increases which eventually results in local melting. once melting starts, that particular region becomes highly conductive, enormous current flow through the material and dielectric breakdown occurs.

### **4. Discharge breakdown:**

Discharge breakdown occurs when the insulator contains occluded gas bubbles. Since gases require smaller ionization potential than solids, the gaseous atom ionize first and the gaseous ions bombarded the solid dielectric causing electrical deterioration.



## 5. Electrochemical breakdown:

Chemical and electro-chemical type of breakdowns are very much related to thermal breakdown. Temperature rise in an insulating material accelerates the chemical deterioration. When temperature rises, mobility of ions increases and hence electro-chemical reaction takes place. When ionic mobility increases leakage current also increases and this may lead to dielectric breakdown.

## 6. Defect breakdown:

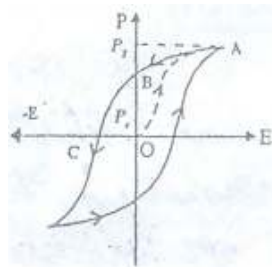
If the surface of the dielectric material has defects such as cracks and porosity, then impurities such as dust or moisture collect at these discontinuities leading to breakdown. Also if it has defect in the form of strain inside the material, that region will also break on application of electric field.

## Classification of Ferro, Piezo and Pyro- electric materials.

### Ferro Electricity:

If a material exhibit electric polarization, even in the absence of external electric field, then these materials are termed as Ferro-electric materials and that phenomenon of exhibiting dielectric polarization, even in the absence of external electric field is known as Ferro-electricity.

These Ferro-electric materials will acquire enormous amount of dipole moment even in a weak electric field and will retain the polarization even when the applied electric field is removed. This analogy can be explained with the help of the hysteresis loop as shown below.



In the figure we can observe that when the electric field( $E$ ) applied the material is increased rapidly from O to A and attains the saturation or spontaneous polarization( $P_s$ ).

Now even when the electric field is reduced to zero (curve A to B), there exists a residual or remanent polarization ( $P_r$ ) denoted by OB.

To remove this residual polarization a negative field has to be applied and that negative field required to reduce the polarization zero is called coercive field (OC).

Thus, the existence of dielectric hysteresis loop indicates that the dielectric materials will have spontaneous polarization, even in a weak electric field.

### **Ferro-Electric materials:**

Ferro-electric materials are classified into three groups, depending on their chemical composition and structure as follows.

#### **(i) Tartrate group:**

##### **Examples:**

1. Rochelle salt ( $\text{NaKC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$ )
2. other salts replaced by  $\text{NH}_4$ , Rb or Ti in the place of Na in Rochelle salt.

#### **(ii) Arsenates and dihydrogen phosphates:**

##### **Examples:**

1. Potassium dihydrogen phosphate ( $\text{KH}_2\text{PO}_4$ )
2. Ammonium dihydrogen phosphate ( $\text{NH}_4\text{PO}_4$ )

**(iii) Perovskite group:**

**Examples:**

1. Barium Titanate ( $\text{BaTiO}_3$ )
2. Lithium Niobate ( $\text{LiNbO}_3$ )

**Properties of Ferro-electric Materials:**

- I. The Ferro-electric materials have permanent dipole moment, even in the absence of external electric field.
- II. They acquire spontaneous polarization even at very weak external electric field.
- III. They exhibit dielectric hysteresis.
- IV. They possess Ferro-electric domain structure
- V. Ferro-electric material will also exhibit piezo-electric effect.
- VI. Ferro-electric materials become para-electric materials above a temperature called Ferro-electric Curie temperature.

**Application of Ferro-Electric materials**

- I. Ferro-electric crystals are used to store energy in small sized capacitors.
- II. They are useful in manufacturing microphones.
- III. Ferro-electric crystals are also used for modulating the signals in optical communication.
- IV. Ferro-electric materials like Electrets will have enormous permanent dipole moment and hence they are used in gas filters, motors, recording units etc.
- V. Electrets are also useful in manufacturing Piezo electric transducers and pyro electric devices such as electric energy converter, infrared detectors etc.

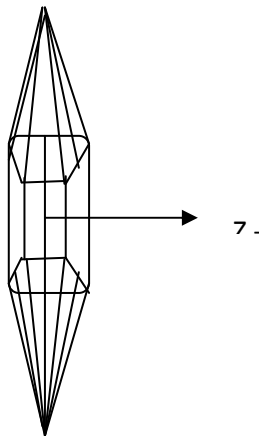
## Piezo electricity:

When certain crystals subjected to compression or tension exhibit opposite kinds of charges at the opposite faces of the crystal. The charges produced on each face are always equal in magnitude and are proportional to the total force applied.

“Materials which are polarized when subjected to mechanical deformation are called Piezo electric materials”

Ferro-electric crystals always exhibit the piezo-electric property. Piezo electric materials are very important since they permit the conversion of the mechanical energy into electrical energy and vice versa. The quartz crystal is a very good examples for this.

Natural quartz crystal has the shape of a hexagonal prism with a pyramid attached to each end. The line joining the end points of these pyramids is called the optic axis or z – axis change in polarization results in change in external field and hence charge on the surface. As it is possible to detect a charge of  $10^{-16}$  Coloumb with a suitable electrometer, temperature changes as small as  $10^{-6}$  C can be measured using the pyro electric effect.



## Pyroelectric materials and their applications

The pyroelectric materials such as  $\text{BaTiO}_3$ ,  $\text{LiNbO}_3$  etc are used to make very good infra- red detectors which can operate at room temperature. Materials such as  $\text{NaNbO}_3$  and other ceramics are used in the construction of pyro-electric image tubes.